some principles of neurobiology—how information is processed in the brain for a specific application like vision. Then I discussed how it can be mimicked on hardware utilising emerging non-volatile memories. We discussed all the other candidates that could be used as neural processing elements, apart from memristors—flash memories, magnetic memory devices, SRAMs, DRAMs, phase change memories, etc. I also talked about all memory technologies that exist right now and evaluated their potential for neuromorphic computing. Moreover, based on the students' feedback for this course, I got a commendation letter from the director for excellence in teaching.

Our NeuroCHaSe research group is also actively working on the development of neuromorphic hardware platforms and ensuring their security against adversary attacks. We are also going to tape out (fabricate) a 3D-neuromorphic computing IC soon. Stay tuned for the news!

Apart from IIT Kanpur, what opportunities exist in India? What does the future look like for Indians?

As of now, the knowledge, dissemination and expertise in this area is limited to the eminent institutes such as IITs and IISc, but with government initiatives such as India Semiconductor Mission (ISM) and NITI Ayog's national strategy for artificial intelligence (#AIforall), this field would get a major boost. Also, with respect to opportunities in the industry, several MNCs have memory divisions in India—such as Micron, Sandisk (WesternDigital), etc—that develop the memory elements which will be used for neuromorphic computing.

There are also many startups working on the development of neuromorphic ICs. One of them is Ceremorphic India Pvt Ltd. They are working on SRAM based neuromorphic computing systems. With the establishment of ISM, I expect more companies to eventually start their neuromorphic teams in India.

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How will it affect normal people and the entire B2B marketplace? Or do we still have time for neuromorphic chips to come to the market?

A lot of companies are already fabricating these chips. Almost all companies have a dedicated team working on neuromorphic computing. These chips are already present in some products as well. If you look at Apple's latest iPhone, they have a chip called M1, which is an AI processor. This is not something that is far off into the future.

In terms of its impact on common people, just consider the field of healthcare. You need to get a lot of data from the patients for the development of personalised medicine, genome sequencing, etc. While the current sensor technology is efficient in extracting such data, processing it in real time is the bottleneck. Neuromorphic computing would enable processing capabilities within the sensors themselves. Therefore, in the future, we may expect significant developments in critical fields like personalised medicine, genome sequencing, etc, fuelled by the massive computing power of neuromorphic hardware.

Not just the healthcare sector, even for drones or warfare systems, the major problem is their processing capability. Their camera captures good quality images but extracting information from the images in real time is the bottleneck. Drones are connected to Wi-Fi and send images to the base station. The base station processes the data and sends the signals back. This is a power-hungry process and can be intercepted too. If the drone itself has the capacity to process the data, it can take immediate action. Just imagine how it can transform the world.

Also, we use all sorts of applications like Alexa, Siri, etc, but they do not work without Wi-Fi. The main problem with these systems is that their processing power is limited. Whenever we give them any input, they just transfer it to the server, where the processing happens. The output comes from the server to the device, and the device simply gives us the output. If you can give such processing power to all the energy-constrained smart devices around you, communication between the server and the device is not needed. In such a scenario, we not only make these devices compute independently but we also reduce the cloud dependence, which is a major source of security vulnerabilities and adversary attacks.

Even in space, rovers capture images and send them back to the earth station for processing. If an energy-efficient processing system is installed there, we'll have Mars rovers and Moon rovers driving on their own. Autonomous driving is not fiction anymore. It just requires an energy-efficient processor to do that, and that is what neuromorphic computing is all about. EFY